



Metal waste in municipal solid waste incineration bottom ash: characterization, source identification, and recovery potential

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Abstract

The steady reduction in landfill capacity and limited land area for constructing new landfills require the recycling of municipal solid waste incineration bottom ash (MSWIBA). While most studies focus on using MSWIBA powder as a construction material, few have addressed metal waste characterization and recovery. This study quantified magnetic metal waste (MW) and nonmagnetic metal waste (NMW), which account for 14.40 ± 4.02 and 0.57 ± 0.20 wt% of MSWIBA, respectively. The identification of original discarded products, combined with elemental analysis, shows that the top 10 MW categories represent over 90% of the total MW weight, and most identifiable MW originated from daily necessities and common metal fittings. Notably, 16.50% of disposed metal waste, which should have been collected through separate collection systems, was found in MSWIBA. Some MW categories contained relatively high Cu, which could affect regenerated steel quality, while trace elements such as Cr, Mn, Co, Ni, As, and Mo were enriched in the magnetic ferrous powder. The estimated amounts of Ni and Co contained in MW landfilled annually correspond to 10.96% and 6.27% of Japan's domestic demand. These findings indicate a significant recovery potential and call for improvements in upstream waste separation and downstream mechanical separation.

Keywords Municipal solid waste incineration bottom ash · Magnetic metal waste · Nonmagnetic metal waste · Recycling potential

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Introduction

Municipal solid waste incineration (MSWI) residue can be divided into bottom ash (BA) and fly ash (FA). FA is collected using a flue gas treatment system and contains high concentrations of heavy metals such as Pb, Cr, and Cd [1–3]. Therefore, from a safety perspective, it needs solidification pre-treatment [4, 5] and is treated as a hazardous waste in many countries. In contrast, BA accounts for approximately 80% of MSWI residues and has a relatively low concentration of heavy metals [6, 7]. Its particle size distribution is similar to that of natural aggregates, making it suitable for reuse as a road pavement and cement additive, particularly in Europe and the USA [8, 9]. However, the practical recycling of BA is still limited because of concerns such as heavy metal leaching [10, 11] and remaining metallic fragments [12, 13]. The estimated service life of landfills in Japan is only 23.4 years [14], their capacity is steadily decreasing, and constructing new landfills is extremely challenging owing to limited land area. Recycling and recovering BA is

still desirable to extend the service life of landfills from the circular economy perspective.

BA contains not only granular ash particles but also metal waste from municipal solid waste (MSW) that has not been separated. Previous studies have demonstrated the presence of valuable metals in MSWIBA. Back et al. found that ferrous and non-ferrous metals accounted for 0.9% and 3.2%, respectively, in MSWIBA < 8 mm from a Japanese MSWI plant [15]. Kaartinen et al. investigated landfilled MSW in Finland and found that approximately 1% of the waste could be recovered as magnetic metal waste via a mechanical process [16]. Kuo et al. also found that some metals identified in Taiwan's MSWIBA had higher concentrations than natural minerals, such as Fe, Al, Cu, and Zn [17]. Pienkoß et al. found that there are 4.64–5.63% of Fe can be separated by wet density separation from MSWIBA and the concentrations increased with particle size [18]. However, most investigations on BA typically focus on the powdery and granular components. Large pieces, including large magnetic metal waste and bricks, were neglected before the investigation, either directly or indirectly [6, 13], resulting in an inaccurate reflection of BA metal recycling potential. According to Huber et al., 50% of the Fe (including magnetic and nonmagnetic ferrous) and 20% of the Cu is distributed in a particle size fraction > 50 mm [12]. The Fe content was substantially underestimated when the large pieces were ignored. Therefore, to more accurately assess the metal recovery potential of MSW streams, further investigations of recoverable metal waste, particularly large magnetic metal waste pieces in MSWIBA, are required.

Japan has a separate MSW collection system intended to divert recyclable metal waste from MSW streams. However, metal waste is still reported to be detected in incoming MSW at MSWI plants [19], indicating that some metal waste is not successfully recovered through source separation. However, previous research did not show which discarded products these metal waste originated from or in what form they remained after incineration. Such information is important for evaluating the effectiveness of existing collection systems. Bunge found a surprisingly large number of metal waste remained in MSWIBA, including bottles, cans, pots and pans, batteries, and small entertainment devices [20]. However, these studies have rarely discussed the characteristics of magnetic metal waste, original discarded products, and their recovery potential. Such information could provide practical guidance for waste separation collection.

Previous studies on metal recovery and reuse from MSWIBA have focused on nonmagnetic metals, particularly Al [21–23]. Magnetic metals are typically ignored because of their relatively mature recovery technologies. However, Fe recovery from MSWIBA is desirable not only for facilitating the reuse of BA but also for recycling Fe resources,

and the characteristics of recovered ferrous metal remain insufficiently understood. The quality of recovered ferrous metals is an important factor affecting their recyclability. Contaminant content will impact recycled steel, causing it to be downcycled to end uses only, which have a greater tolerance for impurities [24]. Regarding scrap from collected industrial waste, Daigo and Dunuwila identified the sources of contamination elements that interfere with recycling [25]. Haupt et al. found that the Fe scrap recovered from MSW in Switzerland was of the lowest quality because of its high Cu and Sn concentrations [26]. In landfill reclamation research, Jain et al. recovered metal waste from excavating waste disposed in landfills, but found that it was of poor quality and unsellable, and terminated the sorting process [27]. Therefore, understanding the elemental composition of recovered metal waste is important for evaluating their recycling potential. A detailed component analysis of recovered Fe is required for assessing their potential reuse pathways.

Therefore, this study aimed to investigate the characteristics of the metal waste contained in MSWIBA to assess their recyclability and feasibility for separate collection. In this study, metal waste was extracted from 300 to 500 kg BA by magnet and visual inspection and analyzed to identify their original discarded products to assess the feasibility of recovery via separate upstream collection. The BA sampling volume was sufficiently large to ensure that large fragments were included to reduce the influence of particle size. Subsequently, the masses and elemental concentrations of the metal waste and residual BA were quantified. Investigation of the contaminant content is helpful for Fe recovery recyclability analysis.

Materials and methods

Materials

In this study, six MSWIBA samples were collected from a stoker-type incineration plant in Kyoto, Japan, equipped with a wet-discharge BA process; one sample was obtained during each of the three sampling events in 2019 and 2020. The material flow in the plant is shown in Fig. 1. Both MSW and bulky waste (e.g., beds and sofas) are treated at the plant. MSW was directly transferred to the refuse pit, whereas bulky waste was shredded by a crusher and passed through a magnetic separator to separate ferrous metals into the hopper. The remaining fraction was conveyed to the refuse pit as combustible waste. The waste in the pit was incinerated. Residues were discharged into the ash pit via a wet conveyor, whereas FA was captured using a bag filter. MSWIBA was transported to a landfill after approximately

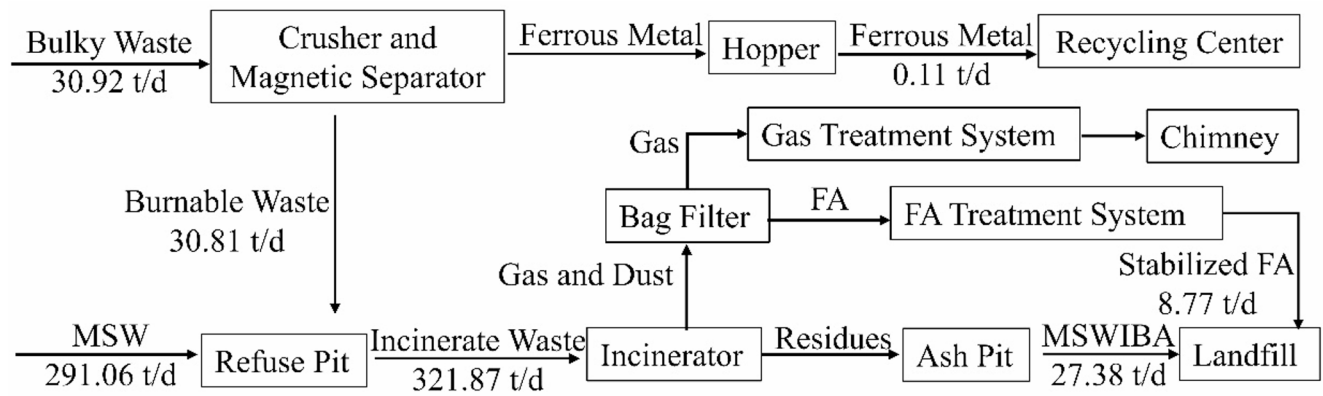


Fig. 1 Material flow of MSWI plant

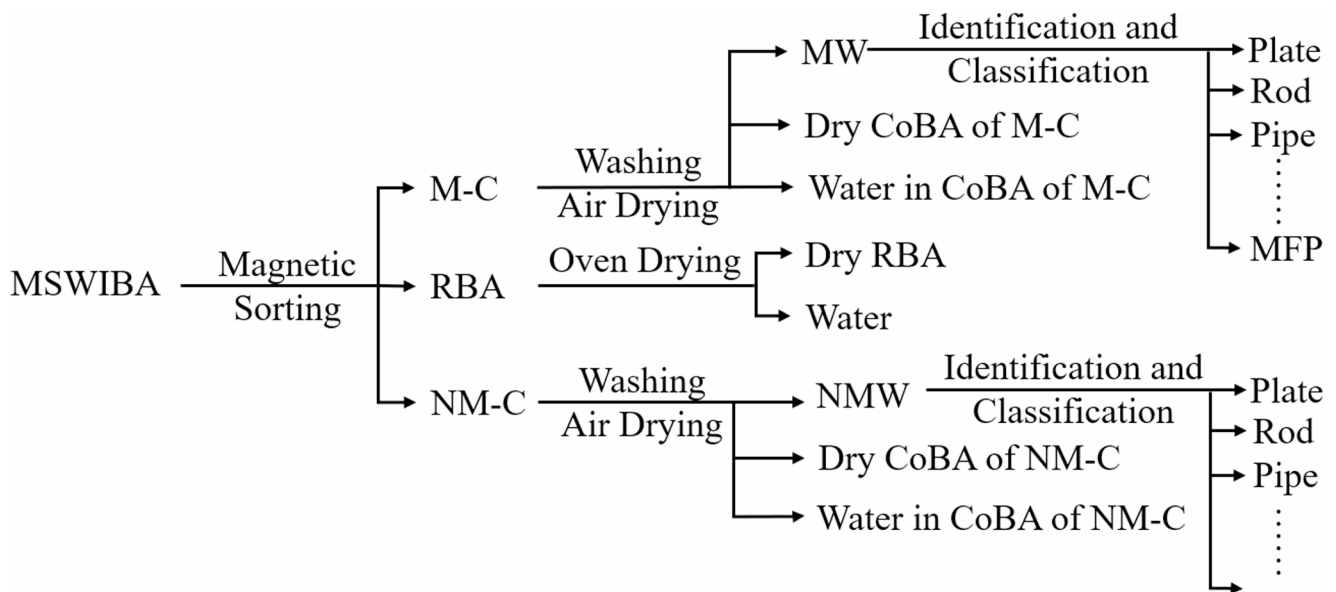


Fig. 2 Sampling process

5 d of aging. FA was stabilized via a treatment system and landfilled.

The sampling process is shown in Fig. 2 and detailed information is shown in Figure S1 (Supplementary Material). For each sampling, 300–500 kg of MSWIBA was collected from the ash pit. A handheld magnet with a magnetic force of 30 N (Trusco Nakayama Corporation, Tokyo, Japan) was used to extract the magnetic component (M-C) together with ash loosely adhering to the surface. All materials that could be attracted by the handheld magnet were included in M-C without predefined particle size range. The total weights of the M-C were recorded. The samples were placed on a sheet, and the adhering ash was removed from surface by washing. After washing, the handheld magnet was used again to extract the magnetic metal without ash, which was defined as magnetic metal waste (MW) sample. The ash remaining on the sheet was collected as co-extracted bottom ash (CoBA). The MW was then air-dried on a clean

sheet for two weeks, followed by visual identification of the original discarded products piece by piece and subsequent piecewise weighing, as described in Sect. [MW and NMW classification investigation](#).

Except for the magnetic materials, materials with a metallic appearance (metallic luster and high density) that were not attracted by the magnet were manually sorted as nonmagnetic component (NM-C) with adhering ash. Similarly, the NM-C samples were placed on a sheet, washed, and then air-dried for two weeks. As this component was sorted visually rather than mechanically, nonmetallic fragments (e.g., pottery or porcelain fragments) may have been mistakenly included. Therefore, a re-identification was conducted to confirm the metal waste, which was classified as nonmagnetic metal waste (NMW) samples and weighed, followed by identification shown in Sect. [MW and NMW classification investigation](#). The metal waste fused with the ash was also included.

After sorting, 15–20 kg of the remaining MSWIBA was sub-sampled via quartering and designated as residual bottom ash (RBA).

Methods

MW and NMW classification investigation

For the RBA, the water content was determined as the weight loss during drying. In this study, 10 g of the RBA sample was dried in an oven at 105 °C for 24 h. The weight loss before and after drying was measured to calculate the water content.

For the MW, the classification was based on the original discarded products, as shown in Fig. 2. The original discarded products were identified based on their appearance (e.g., remaining shapes and structures) and recognizable product characteristics after incineration. When the identification was not possible, the items were classified into six types based on shape and size: plate-like (plate), rod-like (rod), pipe (pipe), lump (> 37.5 mm, lump), and small lump MW (10–37.5 mm, small lump), and magnetic ferrous powder (< 10 mm, MFP). The size thresholds were selected with reference [28]. Each item was individually counted and weighed. The total weight, number, and proportion of each MW type is summarized in the Results and Discussion section. For NMW, classification followed the same criteria as that for MW.

Elementals concentration analysis

The MW and NMW major elemental compositions were determined using portable X-ray fluorescence spectrometry (PXRF; Niton XL3t-950 S, Thermo Fisher Scientific, Billerica, MA, USA) in ore mode. For each MW category, three randomly selected samples were analyzed for their major elements. All NMW samples were analyzed individually.

The RBA, CoBA, and MFP major elemental compositions were determined using X-ray fluorescence spectrometer (XRF; XRF-1800, Shimadzu Corporation, Kyoto, Japan). 100 g of each sample was oven-dried at 105 °C for 24 h and pulverized in a vibration mill for XRF analysis.

The RBA and MFP trace elemental compositions were quantified using inductively coupled plasma mass spectrometry (ICP-MS; XSERIES 2 ICPMS, Thermo Scientific, Waltham, MA, USA) and inductively coupled plasma atomic emission spectrometry (ICP-AES; iCAP7400 DUO, Thermo Scientific, Waltham, MA, USA). For these analyses, 0.1 g of each dried and pulverized sample was digested using a microwave acid digestion procedure. The results were validated using certified reference material (BCR-176,

Joint Research Centre, Geel, Belgium). Additional information is presented in Table S1.

Similar to the leaching behavior analysis provided by Mao [3], 21 trace elements were selected as target analytes. Cd, Se, As, Pb, and B were selected in accordance with Japanese leaching test regulations. In addition, T-Cr, Mn, Co, Ni, Cu, Zn, Mo, In, Ag, Sn, Sb, Te, Ba, Tl, and Bi were selected based on their potential toxicities. The selection was made with reference to other domestic environmental standards in Japan (e.g., wastewater regulations) and international standards for thermal treatment residues, waste, soil, and recycled products. Fe was selected because of its high concentration in MW. All elements were analyzed by ICP-MS, except for Fe, which was measured by ICP-AES because of its high concentration.

All analyses were conducted in triplicate.

Results and discussion

Investigation results

Content results

The MW and NMW contents are shown in Fig. 3. As a result, the average contents of M-C and MW were 21.99 ± 6.94 and 14.40 ± 4.02 wt%, respectively. The CoBA is thought to have the same water content as RBA at $19.64\% \pm 2.25\%$. The MW accounted for an average of 17.30 wt% of the total dry weight. The MSWIBA MW content showed considerable variation across the six samples, reflecting its highly uneven distribution. The volume of the landfill residue can be reduced through MW recovery. The bulk density was estimated at 1.17 and 0.90 t/m³ for MSWIBA and Fe scrap based on operating data, respectively. Considering Japan's estimated landfill service life of 23.4 years, the reduction in landfill residues may extend it to 26.2 ± 0.8 years. The NMW accounted for 0.57 ± 0.20 wt% in total weight and an average of 0.68 wt% in dry weight. As the NMW was re-identified to remove misclassified items, the NM-C content is meaningless. Although the measured NMW content in MSWIBA was <1 wt%, it was likely underestimated because of inherent visual identification limitations.

Metal waste (including MW and NMW) accounted for 19.44 ± 5.55 wt% of dry MSWIBA, of which 96.22 wt% was magnetic. Back et al. reported an Fe metal content of 0.9% in MSWIBA with particle size < 8 mm [15], whereas the largest items in this study were up to about 1 m in size, highlighting the effect of excluding large fragments before analysis.

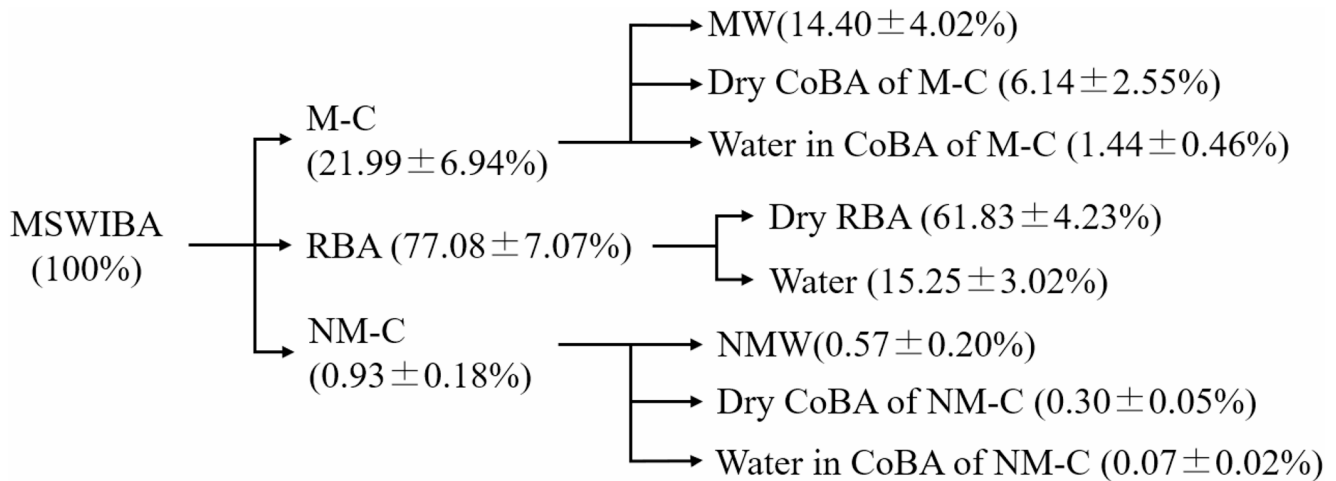


Fig. 3 The content result of MW and NMW ($N=6$)

MW weights and number distributions

The original discarded MW products were identified after air drying. The samples collected from the six sampling campaigns were classified into 88 categories, as shown in Table S2. Among these, 52 categories appeared in at least three sampling campaigns, and 30 categories were present in all six campaigns. Except for six categories whose original discarded products could not be identified, the remaining 24 categories included many types of daily necessities, such as battery, can, lid, hanger, umbrella, spoon, fork, scissors, cutter knife, and various pin and clip categories. Common metal fittings categories—such as screw, strike plate, hinge, jump ring, screw eye, zipper head, and chain—were also frequently observed. In addition to these, the “machine component” category, which includes items used in machines such as motors and toroidal coils, appeared in all six campaigns. Besides the 88 categories, staple pins appeared in large quantities during every sampling event; however, because of their small size and difficulty of complete manual separation, they were not given a distinct classification.

Figure 4 shows MW weight and number distributions across categories. Figure 4(a) shows the MW average weight distribution across categories from six sampling events. The top five categories in the weight distribution accounted for >60% of the total sample weight, and the top 10 categories accounted for >90% of the total sample weight. The top six categories consisted of items with unclear original discarded products, many of which were relatively large that may have originated from bulky waste. The operating data also showed that metal waste accounted for 9.00% of the burnable waste from bulky waste after magnetic separation, as shown in Fig. 1. Assuming no mass change during incineration, this would account for 10.13% of the MSWIBA, nearly 70% of the observed 14.4%, supporting this interpretation.

Figure 4(b) presents the MW average number distribution across categories from six sampling events. Only categories that accounted for >1% of the total number were included. The screw category was the most numerous, accounting for 59.88% of the total. Other high-ranking categories include lid, battery, spring, paper clip, can, hair pin, and other metal fittings categories. This indicates that small metal waste is more likely to be discarded in MSW. Six categories with unidentifiable original discarded products were excluded from the figure because these items may represent fragments of larger objects, making their individual quantities less meaningful. Nevertheless, the number of the plate category was approximately 40% of that of the screw category. The lump, pipe, and rod categories were more abundant than the battery category but less abundant than the lid category.

Bunge found that over 90% of samples were smaller than 100 mm [20]. Despite diligent separate collection efforts, more than half of the small metal consumer products generated by Swiss households still end up as MSW, particularly small metal waste such as clips, batteries, and electronic entertainment devices. The number distribution in this study showed a similar result, indicating that small metal waste is difficult to separate using the separation collecting system and tend to remain in the MSWIBA. However, this study found many items of the plate, rod, pipe, and lump categories were larger than 100 mm, accounting for over half of the weight within each category. The largest metal waste was up to about 1 m in size. The variation in large piece sizes may be due to differences in collection methods between the two countries.

The average unit weights and numbers of the top 10 categories based on weight and number distributions are listed in Table 1. The rod, small lump, and MFP categories were excluded owing to the inability to determine their exact numbers. The rods were entangled and could not be

Fig. 4 The MW weight and number distribution across categories. **(a)** The weight distribution of all items. **(b)** The number distribution of identified items

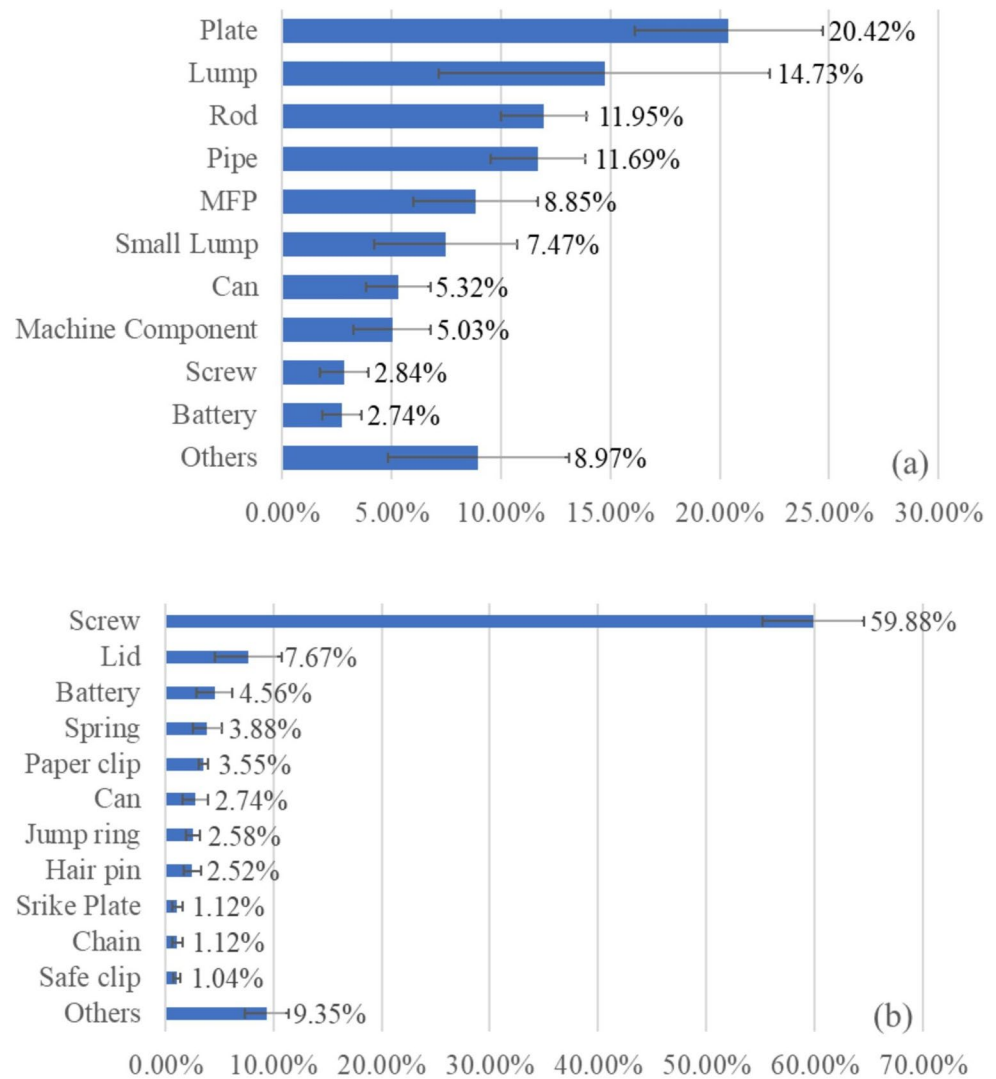


Table 1 Average unit weights and numbers of the top 10 heaviest categories in this study and previous surveys

Category	Plate	Lump	Pipe	Can	Machine component	Battery	Screw
Unit weight (g/item)	30	73	61	67	167	20	1
Unit number (items/t-BA)	1034	313	285	116	42	195	2570
(This study)	±355	±181	±132	±45	±24	±70	±662
Unit number (items/t-BA)				256±33		140	
(Previous surveys)						±14	

separated. The small lumps were too numerous, with an estimated quantity approaching 10 thousand pieces per ton. And the MFP existed in powder form. Regarding unit weights, machine components were the heaviest category, whereas screws were the lightest. The screw category had the highest unit number. Considering they are used to fasten and connect objects; the large number of screws indicates the presence of considerable bulky waste.

Between 2001 and 2006, Kyoto City conducted surveys on the number of cans and batteries in MSWIBA; the results

are presented in Table 1. Compared with previous can survey results, the current study showed a significant decrease. Steel can consumption decrease from 908 thousand tons in 2004 to 328 thousand tons in 2023 [29, 30]. Considering that steel can consumption was more than halved compared to 20 years ago, this reduction may have contributed to the decrease in the number of cans remaining in the BA. In contrast, the results for dry batteries in this study were higher than those in previous surveys. According to the Battery Association of Japan, the sales volume of primary batteries

in 2020 was approximately 60% of that in 2005 [31], indicating a clear increase in the number of dry batteries mixed in MSW. The limited number of specified collection points may increase access distance and reduce the convenience of battery recycling, which can significantly affect recycling rates [32, 33]. A survey also found that only 22.9% of residents disposed of batteries according to the collection rules requiring the use of designated collection points [34], supporting this interpretation.

NMW weights and number distributions

The NMW collected during the six samplings were classified using the same method as the MW after air-drying. Owing to the small sample size compared to the MW, the weight and number distributions were analyzed as totals rather than averages. Figure S2 presents the distribution results across categories. Of the 88 categories predefined in the MW analysis, 25 were identified. As with the MW, items of unclear original form were grouped into the plate, rod, pipe, small lump, and lump categories. In addition, categories related to daily necessities (10) and metal fittings (6) were identified in the NMW samples. The predominance of daily necessities and metal fittings may be because familiar shapes are easier to recognize and select during visual sampling. As shown in Figure S2, the lump, plate, can, pipe, rod, spoon, and lid categories ranks were at the top of both weight and number distributions. A surprisingly large number of coins was found in the Swiss MSWIBA nonmagnetic metal fraction, approximately half of which were foreign currencies [20]. This contrasts markedly with the current study, in which no coins were identified across the six sampling campaigns in the NMW. This discrepancy may be explained by the differences between Switzerland and Japan in terms of geographical location and neighboring countries.

Recycling potential before incineration

According to the laws and regulations, spray cans and metal waste within a certain size constraint should be collected as “small metal items” on a monthly schedule. Similarly, household appliances within certain size constraints were recycled as “small household appliances” at a few specified collection points. These items were directly transferred to the recycling plant. Items that exceed dimensions limitations should be disposed of as bulky waste at additional cost and passed through the crusher and magnetic separator shown in Fig. 1. However, although small daily necessities such as clips are impractical to separate in daily life, many waste that should have been sorted in advance still remained among the 88 categories. They are machine component, battery, can, hanger, spoon, fork, scissor,

cutter knife, cell battery, mesh shelf, wrench, utility tong, butter knife, plier, golf club, scraper, bottle opener, hammer, spade, ladle, hedge shears, pot, dustpan, hoe, whisk, bucket, handsaw, vacuum flask, steaming plate, grater, udo ladle, alarm clock, and shaver categories. These categories together accounted for 16.9 wt% of MW and 6.5 wt% of NMW, respectively. Based on the contents of MW (14.4 ± 4.02 wt%) and NMW (0.57 ± 0.20 wt%) shown in Fig. 3, the additional recyclable amount of metal waste that could have been separated in advance was estimated as a weighted sum of their respective contributions, resulting in 16.50 ± 0.16 wt%. The results show that compared to disposing of these waste in accordance with relevant regulations, placing them in combustible waste appears to be a much simpler option for the public. Although current recycling policies and separation and collection systems have successfully enabled the recovery of a substantial proportion of metal waste [35], further efforts are required to improve the effectiveness of separate waste collection through education and awareness programs. To improve recycling, measures are required to support the policy, such as increasing the number of collection points and the frequency of door-to-door pickups, reducing waste disposal fees, and enhancing the early separate collection of these waste. In addition, AI-based waste identification and sorting technologies have shown strong potential as a pathway for advancing sorting efficiency in future recycling systems [36, 37].

Elemental content analysis

MW elemental content

Based on MW weight distribution across categories, the top 10 categories determined before were further analyzed for surface elemental concentrations. The average results are shown in Table 2, and supplementary information is presented in Table S3. Elemental contents below detection limit of 0.1 wt% are denoted by “/”. The PXRF results reflected the surface compositions affected by oxidation during incineration and cooling. The Fe content across all MW categories was relatively consistent, with an average concentration of 47.5 ± 1.4 wt%, indicating a high degree of surface oxidation. Notable differences among the categories were observed in the Mn and Zn concentrations, which were relatively elevated in the battery category, with average contents of 1.0 ± 0.3 and 1.2 ± 0.2 wt%, respectively. This is consistent with the widespread use of alkaline batteries, which contain manganese dioxide (MnO_2) and Zn. In 2022, approximately 2.83 billion primary batteries were sold in Japan, of which 44.2% were alkaline [31].

The Cu content in the rod and screw categories was relatively high, reaching 4.2 ± 2.4 and 2.3 ± 0.2 wt%,

Table 2 The MW elemental concentrations on the surface (wt%) *

Fe	Plate	Pipe	Rod	Lump	Small lump	Machine component	Battery	Screw	Can	MFP	Average
	50.0	48.7	39.9	48.2	42.0	44.1	44.4	44.4	61.4	50.1	47.5
Mg	0.4	0.5	0.3	0.3	0.4	1.0	0.4	0.2	0.3	/	0.4
Al	0.3	/	0.1	0.1	0.2	0.2	0.1	/	0.4	0.1	0.2
Si	0.1	/	0.2	/	0.5	/	/	/	0.2	/	0.1
S	0.2	0.2	0.3	0.2	0.3	0.3	0.2	0.2	0.3	0.2	0.2
Cl	0.5	0.1	0.3	0.2	0.3	0.2	0.1	0.1	0.5	0.1	0.3
K	1.0	2.5	0.4	3.0	0.6	2.5	6.3	0.3	0.5	2.7	1.8
Ca	2.0	1.3	4.1	4.1	7.5	3.2	3.3	3.3	3.9	3.8	3.4
Ti	0.7	0.2	0.2	0.1	0.3	0.1	0.1	0.1	0.2	0.1	0.3
Cr	0.1	0.1	0.1	0.1	0.4	0.1	0.2	0.1	0.1	0.1	0.1
Mn	0.3	0.1	0.2	0.1	0.2	0.7	1.0	0.2	0.2	0.7	0.3
Co	/	0.1	/	0.2	/	0.2	0.3	/	/	0.3	0.1
Ni	4.5	8.2	/	3.3	/	1.2	0.4	1.2	/	1.4	2.8
Cu	0.1	/	4.2	0.4	0.7	0.1	0.1	2.3	0.1	0.1	0.8
Zn	0.6	0.2	0.3	0.2	0.3	0.4	1.2	0.3	0.1	0.9	0.4
Others	39.0	37.2	49.2	39.4	45.5	45.7	41.7	47.1	31.5	38.5	41.3

*: $N=18$ for each category (6 samplings $\times$ 3 replicates).

respectively. In Fe recycling systems, steel scrap is typically melted in electric furnaces to produce new products. Therefore, elements that vaporize after heating are unlikely to affect the products. However, those that remain in the products as alloy metals, such as Cu and Sn, can easily interfere with the quality of new Fe production [38]. High Cu and Sn concentrations reduce the recyclability of steel scrap because these elements are considered critical impurities. Specifically, when Cu and Sn concentrations exceed 0.1 and 0.04 wt%, respectively, they can cause hot shortness, leading to surface cracking during hot working processes [39]. According to ISO standards, the maximum allowable Cu content in carbon steel is 0.4 wt%, and for reinforcement bars is 0.89 wt% [40]. The results of this study indicate that rods and screws are not suitable for recycling and that lumps and small lumps can only be recycled for reinforcement bars. A similar issue has been reported in Switzerland: the Fe scrap recovered from the BA of MSWI plants often must be exported for quality reasons [26]. Thus, the high Cu concentration in steel scrap is recognized as a global challenge for sustainable metal recycling. However, considering that the average grade of Cu ore is decreasing and currently ranges between 1 and 2 wt% [41], some waste may possess significant recycling value as a secondary Cu resources. Among the machine component analyzed, some—such as coils—were found to contain Cu wire wound around the core structure. The partial Cu content of the Cu wire reached 82.0 wt%. In recent years, Cu recovery from scrap wire has attracted increasing attention [42]. To increase the value of material recovery, it is important to improve the disassembly process and sort different components, such as Fe and Cu wire, for separate recycling.

NMW elemental content

The NMW elemental contents, based on all collected samples, is summarized in Table S4. Only elements with concentrations higher than detection limit (0.1 wt%) are included. Owing to PXRF limitations in ore mode, elements with atomic numbers <12 cannot be detected. The results show that Fe accounted for 27.26 wt% of NMW, followed by Cu (6.78 wt%), Cr (6.07 wt%), Ni (4.65 wt%), Ca (4.18 wt%), Zn (2.23 wt%), and Al (1.25 wt%). The NMW elemental content, classified by the element with the highest concentration, is shown in Fig. 5; only elements >0.1 wt% are included. Five elements, namely Ca, Al, Fe, Cu, and Ni, were identified as the predominant elements in one or more metal waste items. Based on the highest elemental concentration, the NMW samples were categorized into five groups: Cat-Ca, Cat-Al, Cat-Fe, Cat-Cu, and Cat-Ni, with average unit weights of 23.83, 23.69, 39.63, 41.35, and 243.95 g per item, respectively. The results showed that Ca was the dominant element in some items, which were likely metal fragments that were molten and incorporated into the ash.

The Ca concentration in Cat-Al group was also high. The melting point of Al (660 °C) is lower than the typical temperature of an incinerator (850 °C), suggesting that Al melts during incineration and subsequently forms aggregates by bonding with other metals. Saffarzadeh et al. found that if Fe and Si were present in the materials, the liquidus temperature of the Al melt system was estimated to be lower than the temperature inside the incinerator [43]. Common sources of Al in MSW include beverage cans, foil containers, and thin foils [44]. In Japan, separate MSW collection has been implemented. In 2023, the demand for Al cans

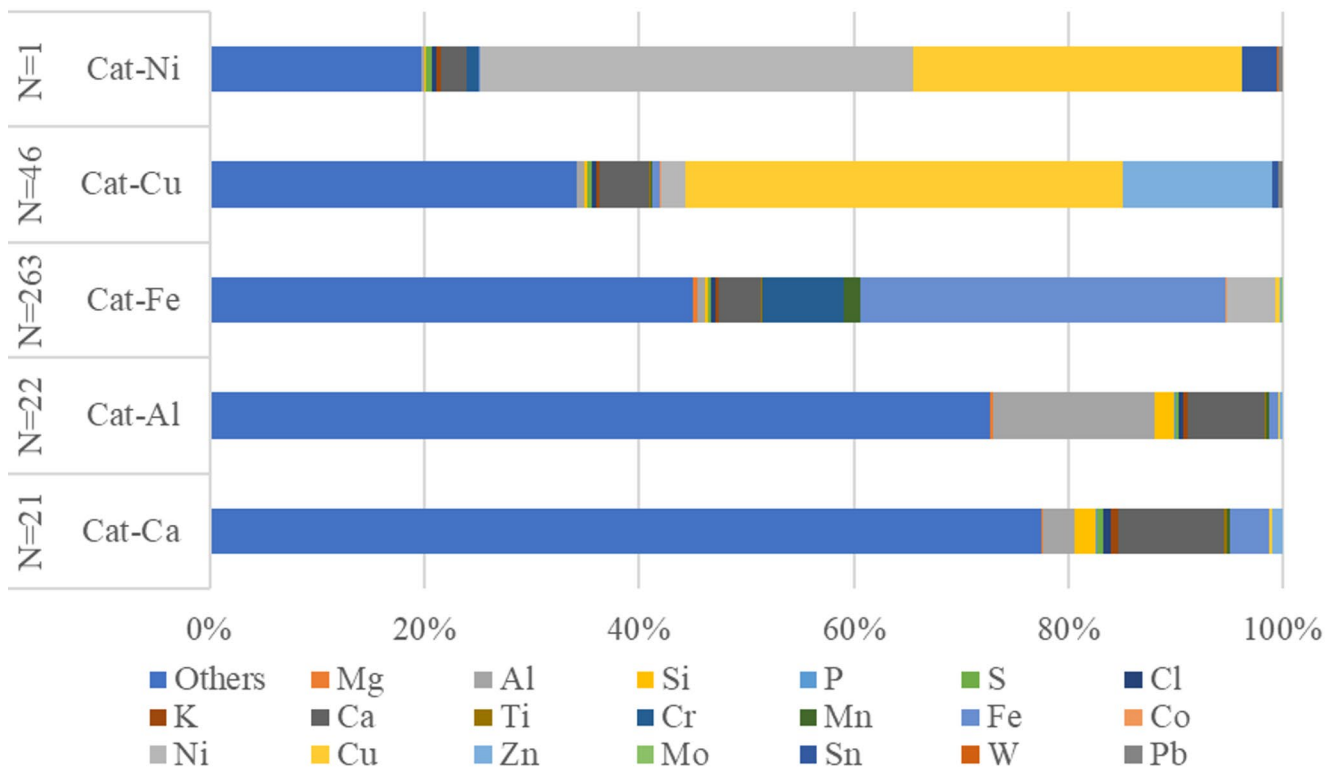


Fig. 5 The NMW elemental content averages across five categories

was 314 thousand tons, with a recycling rate of 97.5% [45]. Assuming that all unrecycled Al cans were transported as incineration waste, this would amount to only 7,866 tons. In contrast, the average annual waste generation per capita in Japan is approximately 400 kg, of which 0.6 kg is Al thin foil [46]. Given that the total amount of MSW incinerated in Japan in 2023 was 30.24 million tons [47], the estimated quantity of Al thin foil incinerated would be approximately 45,360 tons—significantly exceeding the weight of Al cans, suggesting the possibility that a substantial proportion of the Al contained in MSWIBA originated from thin foil. The relatively low unit weight of Cat-Al is also consistent with this inference.

The Cat-Fe group was the most abundant, with 263 items, accounting for 78.09 wt%. Within this category, Cr and Ni were the second- and third-most abundant elements, respectively. Excluding unidentified items, kitchenware represents the most numerous and heaviest category within the classified NMW. SUS304 and SUS316 are the most commonly used stainless steels in the food and beverage industry [48], both of which contain over 50 wt% Fe with substantial Cr and Ni contents. This composition is consistent with the observed elemental concentration patterns, indicating that stainless steel is the primary source of Fe, Cr, and Ni in the NMW. In addition to Cr and Ni, Mn also exhibited a relatively high concentration, indicating a contribution from Mn steel. For the Cat-Cu group, Zn was the second most

abundant element, indicating that it originated from brass alloys. In addition, Mo was detected only in the Cat-Fe group, suggesting it originated from stainless steel.

Based on the inferred physical properties associated with these material sources, NMW is considered suitable for sorting using existing separation technologies, such as combined gravity-based and eddy current separation [49].

Powdered BA elemental composition

The samples used in the elemental composition analyses were RBA (the remaining BA after magnetic sorting), CoBA (ash loosely adhering to the MW), and MFP (MW with particles < 10 mm). The average primary constituent analyses results for RBA, CoBA, and MFP are shown in Figure S3. The trace elemental concentrations of the RBA and MFP are summarized in Table S5 with their detection limits. The elements that exceeded 1,000 mg/kg in both materials were Mn, Cu, Zn, Ba, and Fe. Compared with RBA, MFP showed higher Fe, Cr, Mn, Co, Ni, As, Se, Mo, Cd, In, Sn, and Bi concentrations, whereas Cu, Zn, Ag, Sb, Ba, and Pb were enriched in RBA. The MFP/RBA concentration ratio indicated the enrichment tendency of each element. Cr, Mn, Co, Ni, As, Se, Mo, and Sn exhibited ratios > 2, clearly indicating a tendency to accumulate in the MFP. Cu and Zn are diamagnetic; Co and Ni are ferromagnetic; Cr and Mn are widely used in steel production; and Sn is often used in steel

plating. This behavior is consistent with their adhibition. In addition, Se and Mo enrichment mechanisms remain unclear and require further investigation.

BA elemental recovery potential

In this section, dry MSWIBA was divided into RBA, MW, and NMW. Considering their similar characteristics, the dry CoBA shown in Fig. 3 was calculated as the RBA in the elemental distribution. Owing to its unique properties, the MFP was listed separately from the MW in the elemental distribution analysis. The BA elemental distribution, based

on the dry weight after washing and air-drying, is shown in Fig. 6 with their values shown in Table S6. As a further analysis element, Se was detected only in MFP at trace levels, while levels in RBA, MW, and NMW were too low to quantify. Therefore, Se distribution in different sample fractions is not shown in Fig. 6. RBA accounted for 80.56 wt% of the total dry weight, leading to enrichment of most of the elements in this fraction. The elements distributed in RBA with <50 wt% were Fe (36.91 wt%), Cr (35.92 wt%), Ni (3.37 wt%), Co (29.76 wt%), As (40.53 wt%) and Mo (24.24 wt%). MW excluding MFP accounted for 17.10 wt% of the total weight and contained 56.12 wt% of Fe, 90.48

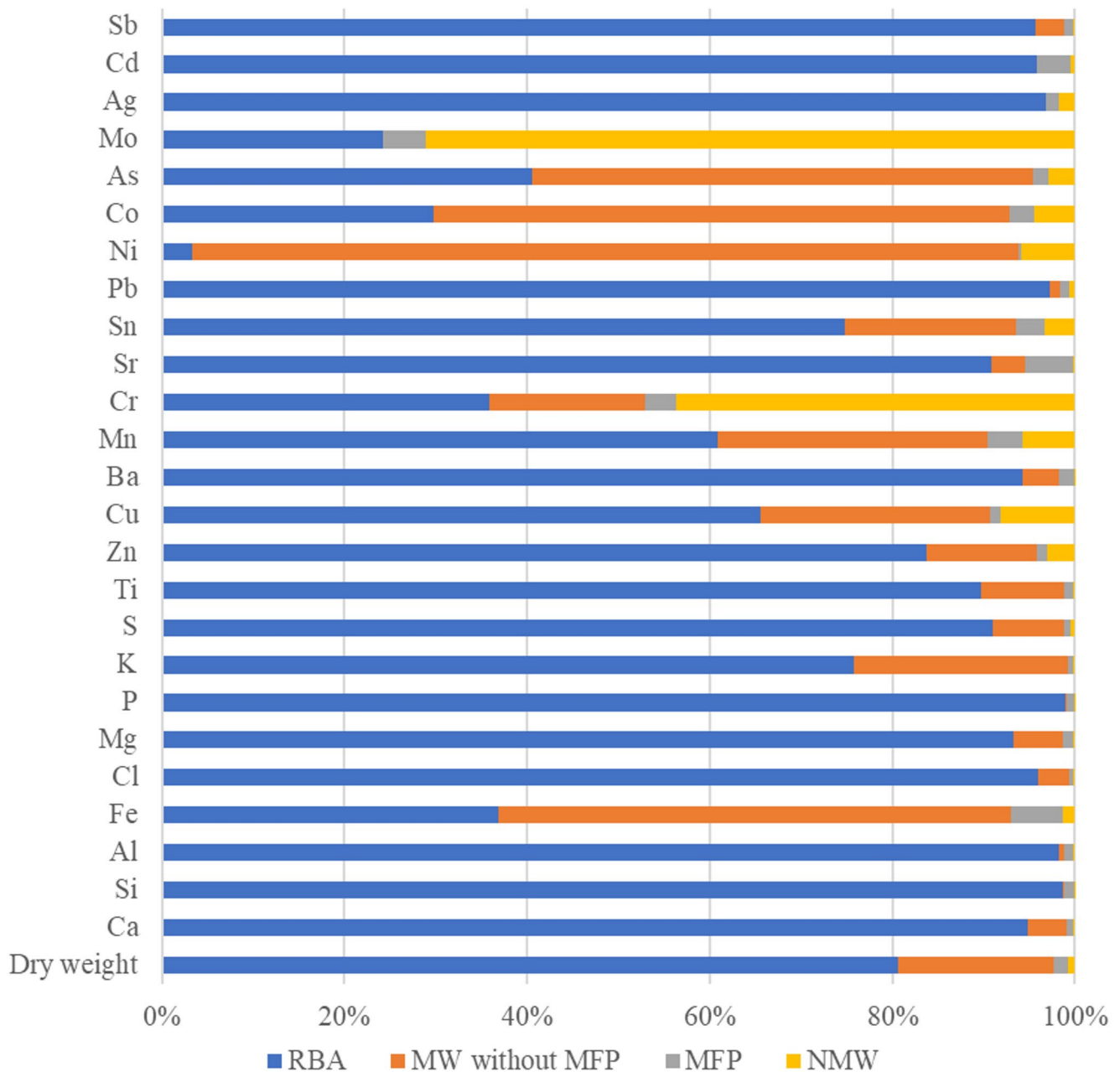


Fig. 6 The MSWIBA elemental distribution in different fractions

Table 3 Estimated recovery potential in MSWIBA (tons per year)

Weight	MW	NMW
	127,057 ± 6,844	4,988 ± 269
Fe	60,064 ± 3,236	1,316 ± 71
Mg	507 ± 27	19 ± 1.0
Ti	367 ± 20	6.8 ± 0.4
Zn	464 ± 25	109 ± 5.9
Cu	1,098 ± 59	327 ± 18
Ba	40 ± 2.2	0.7 ± 0.0
Mn	333 ± 18	60 ± 3.2
Cr	124 ± 6.7	292 ± 16
Sr	8.1 ± 0.4	0.3 ± 0.0
Sn	40 ± 2.2	6.6 ± 0.4
Pb	9.3 ± 0.5	4.6 ± 0.2
Ni	3,804 ± 205	224 ± 12
Co	108 ± 5.8	7.0 ± 0.4
As	7.0 ± 0.4	0.3 ± 0.0
Mo	0.0	4.6 ± 0.2
Sb	3.0 ± 0.2	0.2 ± 0.0

wt% of Ni, 63.07 wt% of Co, and 54.94 wt% of As. Given that the MFP had a relatively low mass content of 1.61 wt%, no elements in the MFP exceeded a distribution of 10 wt%, and only Fe (5.63 wt%) and Sr (5.29 wt%) exceeded 5 wt%. The Fe content of MW included MFP reach 61.75%, which could be considered as the recovery efficiency of the magnetic separation.

The annual volume of landfilled incineration residue in Japan in 2020 was 2.82 million tons (wet weight), of which 1.58 million tons was BA and generated by stoker-type incineration plants [47]. Based on the results of this study, the recovery potential of the elemental fraction in Japan was preliminarily estimated. However, the investigated facility ranked at the 89.9th percentile nationally based on the average noncombustible waste ratios of facilities for 2019–2020, suggesting a higher metal waste content than in most facilities. To avoid overestimation, metal waste was assumed to be proportional to noncombustible waste ratio. The estimated metal recovery potential was adjusted using the ratio of the mean noncombustible waste ratio derived from the fitted Gamma distribution (Figure S4) to that of the investigated facility, with uncertainty represented by the corresponding 95% confidence interval. The estimated recovery potentials of the metals with relatively high values are summarized in Table 3. The results suggest that approximately 127 thousand tons of total MW may be disposed in landfills annually in Japan, potentially containing 60.1 ± 3.2 thousand tons of Fe, 3.8 ± 0.2 thousand tons of Ni, 108 ± 5.8 tons of Co, and 7.0 ± 0.4 tons of As. These results indicate that magnetic separation system may be effective and efficient to recover these elements, particularly Ni, from MW. Compared to Japan's Fe scrap recycling volume of 25.43 million tons in 2023 [50], the estimated MW recycling

potential in BA accounted for approximately 0.50%. And compared to Japan's domestic Ni demand of 34.7 thousand tons in 2020 [51] and Co demand of 1,728 tons in 2020 [52], these amounts represent 10.96% and 6.27% of the total, respectively. Note that Co demand in cathode materials for lithium-ion batteries is not included in the statistics due to the lack of available data. The results suggest that the MW could constitute a substantial potential resource for recovery if effective separation is achieved.

NMW accounted for 0.73 wt% of the total weight and contained 43.66 wt% of Cr and 71.08 wt% of Mo. As shown in Sect. **NMW elemental content**, both Cr and Mo were abundant in NMW, particularly in items classified as Cat-Fe, likely originating from stainless steel. Therefore, there is significant potential for recovering Cr and Mo through stainless steel recovery. However, the most common metal recovery systems for MSWIBA, magnetic, and eddy current separators, cannot effectively separate stainless steel, which represents a component of the NMW [20]. In Vienna, inductive sorting systems are employed for NMW recovery [53], and one company in Japan has begun similar recovery efforts. According to the estimated results of this study, 4.99 ± 0.3 thousand tons of NMW are landfilled annually in Japan, potentially containing 292 ± 16 tons of Cr and 4.6 ± 0.2 tons of Mo. Compared to Japan's Cr consumption of 380 thousand tons in 2022 [54] and Mo demand of 22 thousand tons in 2020 [55], these amounts represent 0.08% and 0.02% of the total, respectively. Although relatively small, these quantities indicate that NMW represents a potentially recoverable resource.

Conclusion

This study characterized the metal waste contained in MSWIBA produced through a wet-discharge BA process with a focus on MW and NMW. MW (14.4 ± 4.02 wt%) and NMW (0.57 ± 0.20 wt%) constituted non-negligible fractions of MSWIBA, indicating that effective metal recovery could substantially extend landfill service life and promote resource recirculation.

Unlike most previous studies that focused only on elemental concentrations, this study also identified the original discarded product sources of metal waste, showing that most identified metal waste originated from daily necessities and common metal fittings. The diversity of small metal waste items in MSWIBA indicates that the current separation and collection systems generally struggle to capture finer items. The recyclable metal waste equivalent to 16.50 ± 0.16 wt% should have been collected through separate designated collection systems in advance, however, they were not. The increased presence of dry batteries compared to 20 years

ago further indicates deficiencies in current waste-sorting behavior and infrastructure. These findings highlight the requirement for improved public participation, education, and more accessible collection points. Strengthening policies and collection systems for small metal items and small household appliances can significantly enhance upstream metal recovery efficiency. In addition to MW, NMW appears suitable for separation using dedicated sorting systems in treatment facilities following incineration based on its inferred composition.

Elemental analysis showed that certain MW categories, particularly rods and screws, exhibited at relatively high Cu content, reflecting a commonly reported challenge in the sustainable metal recycling of multi-metal components that are mechanically inseparable. Improving the dismantling strategies to separate Fe- and Cu-bearing components can enhance recycling efficiency. Trace elements such as Cr, Mn, Co, Ni, As, Se, Mo, and Sn, were preferentially enriched in the MFP, potentially compromising the quality and mechanical performance of the regenerated steel. Large quantities of MW and NMW are still landfilled annually, leading to the loss of metals such as Fe, Ni, Co, Cr, and Mo. Therefore, the recycling of metal waste to recover these metals is justified by both resource recovery benefits and environmental risk reduction.

Overall, the results indicated that increasing the application of existing mechanical recovery processes and improving upstream waste-separation practices are necessary for effective metal recovery from MSWIBA. However, further research is required to evaluate the economic feasibility and environmental impacts of metal recovery using a life-cycle assessment framework, as well as the influence of discharge conditions on metal oxidation and recovery efficiency.

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